Systems of distributed artificial intelligence for analysis of oil product transportation processes: evidence from Russia

Elena Serova* and Daniil Shklyaev

HSE University,
3 Kantemirovskaya St.,
St. Petersburg, 194100, Russia
Email: egserova@hse.ru
Email: dshklyaev@hse.ru
*Corresponding author

Abstract: At present, the use of artificial intelligence (AI) methods and tools is an essential component of management information systems for a company to succeed in a rapidly changing environment. Agent-based modelling systems as systems of distributed AI must be considered nowadays as an obligatory stage of decision-making in Russian oil and gas companies, which use modern information technologies actively. The paper is focused on the description and comparative analysis of system dynamics and agent-based modelling, used for intelligent decision support systems development in transport logistics. The main goal of this research is evaluation of the multi-agent system's role for decision-making processes and management information systems development and creating the model of logistics processes (the processes of oil and oil products transportation, loading, and unloading). The work is based on a generalisation of theoretical researches in this area along with international practices and domestic experience.

Keywords: artificial intelligence; distributed artificial intelligence systems; simulation; agent-based modelling; transport logistics; the processes of oil and oil products transportation; Russia.

Reference to this paper should be made as follows: Serova, E. and Shklyaev, D. (2021) 'Systems of distributed artificial intelligence for analysis of oil product transportation processes: evidence from Russia', *Int. J. Simulation and Process Modelling*, Vol. 17, Nos. 2/3, pp.106–115.

Biographical notes: Elena Serova is an Associate Professor working now in the National Research University Higher School of Economics St. Petersburg Branch, Russia. Her role combines teaching and research in equal measure. She also is the Academic Supervisor of PhD students Doctoral Schools of Economics and Management. Her research interests are related to artificial intelligent, intellectual data analysis, business analysis, strategic management, marketing, and information management. She has co-authored the books and contributed chapters to several books and collections of essays, regular key presenter at national and international conferences and workshops. As a research-active academic with several PhD students under her supervision, she is focusing on complex information systems modelling, marketing information systems, and business models in global environment.

Daniil Shklyaev is currently a Doctoral student at National Research University Higher School of Economics in St. Petersburg [Program 'Economics and Management of the National Economy (Logistics)']. Research theme: 'Comparative analysis based on simulation modelling for oil products logistic schemes of loading and transportation'. He regularly participates in scientific conferences. He works at PJSC Gazprom Neft as the Chief Specialist of the Development Section of the Logistics and Transport Department. In 2017, he is the winner in the tournament 'League of profesSionals' of PJSC Gazprom Neft, which was held among the logisticians of PJSC Gazprom Neft. In 2019, he won the 'Best Expert' nomination as part of the Gazprom Neft tournament 'League of Professionals'. His research interests are transport logistics, development of logistics infrastructure, modelling, and simulation.

This paper is a revised and expanded version of a paper entitled 'Agent-based modeling for decision-making support: case of transport logistics in oil company' presented at the 32nd European Modeling & Simulation Symposium (I3M, EMSS 2020), 16–18 September 2020, virtual.

Copyright © 2021 Inderscience Enterprises Ltd.

1 Introduction

Understanding the exceptional role of intelligent technologies and systems in the new digital economy has given rise to another technological race – for artificial intelligence (AI). In the past few years, two dozen countries have launched their national strategies in the field of AI. Many countries have already developed their AI strategy at the official level during the last three years (Dutton, 2018). The leaders in this list are the USA, China, France, the UK, Germany, South Korea, Japan, and European Union. In Russia, the Decree of the President of the Russian Federation on the Development of AI was launched in 2019 (Serova, 2020a, 2020b).

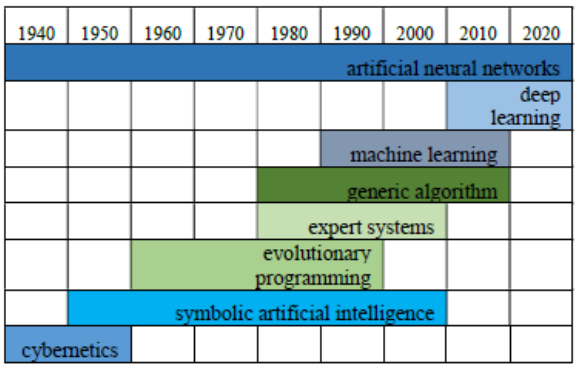
Researchers highlight the following key areas of AI development:

- Predictive big data analytics.
- Ontologies/knowledge management.
- Optical character recognition (OCR).
- Natural language processing (NLP).
- Speech recognition, speech-to-text (STT), linguistic analysis.
- Expert systems, recommender systems, collaborative filtering, information, and analytical systems, design, and management automation.
- Specialised processors and computing systems for AI.
- Image processing, computer vision, classification of images.
- Robotics, smart machines.
- 'Smart' grids and systems in energy, communications, urban economy and other industries, 'smart home', 'smart city'.

Figure 1 shows the evolution of AI development: from cybernetics to machine learning (ML) and reinforcement learning. ML has been actively developing since the late 90s until recently, for more than 20 years. ML provides an opportunity for continual assessments of data to detect and analyse the anomalies and nuances to improve the precision of models and rules. Computer vision produces numerical or symbolic information from images and high-dimensional data. Since 2015, the number of studies related to AI and the obtaining of qualitatively new results through the use of neural networks consisting of many layers has increased. Problems solving with the help of such neural networks is called deep learning. At present, major progress in the use of neural networks occurs in the fields of computer vision, natural language word processing, and reinforcement learning.

There are some tendencies in the sphere of AI development. First of all, interest in the topic of artificial general intelligence (AGI) has increased. In 2019, Microsoft invested $1 billion in OpenAI to support AGI. On the second, cloud AI is evolving rapidly.

Figure 1 AI: the brief history of the development (see online version for colours)

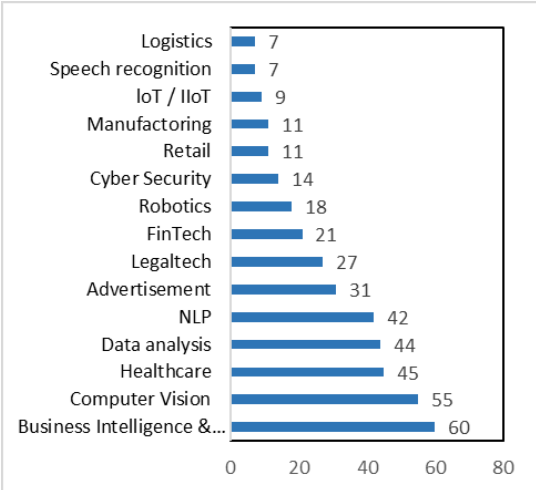


Source: Adapted from Yandex (2019) and Smith et al. (2006)

The quality of basic models improves, particularly in cognitive services such as machine translation, image classification, etc. Also, more trainable models appear that do not require a large amount of data or deep expertise from the user, for example, Google AutoML. Tools for developing and training models are simplified: cloud notebooks like Jupyter/Colab, distributed training of the models, models and components repositories, and other solutions.

According to IDC, in 2019 the AI market in Russia amounted to $139.3 million (ABBYY IDC Research, 2019). Russia is among the top 25 countries in terms of the total number of patents, publications, and abstracts of scientific conferences on AI (Yandex, 2019). However, its lag behind the leading countries such as the USA and China is significant.

Figure 2 Map of the Russian AI companies: technologies used and main activities (see online version for colours)



Source: Adapted from AiRussia Online (2020)

In 2020, about 400 companies were working in the field of AI in Russia. In Figure 2 Russian AI companies on the map are divided by technologies used and main activities. The main players in terms of revenue (over 50%) are several large companies, such as Yandex (≈ 30%), Mail.Ru Group,

Sberbank Technologies, Croc, Kaspersky Lab, Avito, MTS, Yota, etc. The main feature of these Russian companies is that AI has a significant impact on their revenue and makes a significant contribution to their business model.

According to Serova (2020c), intelligent technologies add to the globalisation of business by providing quick access to employees, customers, and partners worldwide, as well as coordinating global interaction between companies at different stages of the value chain. It does not mean that intelligent technologies and systems simply increase the efficiency of a company's operations; they can be considered as a key intangible asset. AI can be considered as a set of technological solutions that are at the stage of active development, and it has already been successfully used in many industries, economic sectors, and companies. AI techniques include data mining, database, ML, pattern recognition, and knowledge discovery (Serova, 2020c).

When decision making on the strategic level of management, the use of distributed artificial systems such as multi-agent systems (MAS), can be considered now as an effective tool for performance improvement. Intelligent agents (intellectual agents) – programs that independently perform tasks that are specified by the user or by other programs, as a rule over extended periods, and are used for data collection and analysis. According to Yilmaz (2020), the ability to systematically vary and experiment with multiple model features is paramount for problems that involve uncertainty and representational ambiguity.

Currently, in the context of the development of the paradigm of the digital economy around the world, there has been a significant increase in interest in the use of intelligent methods and systems of distributed AI, the implementation of appropriate information and communication technologies in organisations which are seeking to ensure a stable economic situation in an aggravated competitive struggle. When decision making on the strategic level of management, the use of distributed artificial systems such as MAS, can be considered now as an effective tool for performance improvement. Intelligent agents (intellectual agents) – programs that independently perform tasks that are specified by the user or by other programs, as a rule over extended periods, and are used for data collection and analysis. The involvement of information technologies and procedures, as well as modern simulation systems, in the first place, MAS – systems of distributed AI, allows you to bring the solution of logistics problems to quantitative results, which are especially valuable for managers.

The study of problems that occur in the development of the theory and practice of logistics is carried out taking into account the application of a systematic approach and modern technologies, such as intelligence data analysis and simulation. Along with others, such a logistics subsystem as transport logistics is especially important. The functionality of transport logistics is actively used in various sectors of the economy. For example, vertically integrated oil companies create complex supply chains. Methods and means of transport logistics are used to deliver oil from the field to refineries, then for loading, and subsequently for transportation and discharge of various oil products to numerous consumers in the Russian Federation and abroad.

The purpose of this work is evaluation of the role such distributed AI systems as MAS for decision-making processes and management information systems development and creating the model of logistics processes (the processes of oil and oil products transportation, loading, and unloading). The paper also focuses on comparative analysis of such approaches in simulation as agent modelling and system dynamics (SD), as applied to logistics processes (the processes of oil products transportation, loading, and unloading). To achieve this goal, it is necessary to develop principles for constructing a model for the process of unloading and loading oil products in AnyLogic software.

Modern research claims that decision-makers operate in a complex and rapidly changing environment, which is characterised by the fact that:

It is necessary to make several decisions to achieve the goal, each of which is considered in the context of other decisions.

- Decisions are not independent; each subsequent decision is limited by the consequences of previous decisions and in turn imposes restrictions on the following decisions.
- The decision-making environment changes on its own, as well as a result of decisions.
- Decisions are made in real-time.

The methodology of the research is based on a system approach, comparative analysis, and methods and procedures of modelling.

2 Theoretical fundamental and brief literature review

The foundations of decision-making research in a dynamic environment were laid by Forrester (1961). Further, Pospelov (1998), Sterman (2000), Law and Kelton (2000), Wooldridge (2002), Pidd (2004) and Albright et al. (2011) made their contribution to this area.

AI and the technologies that support it, such as big data, cloud, internet of things (IoT) sensors, and ML, are becoming researchers, collaborators, augmented assistants, and even proxies for humans in the customer experience (SAP SE, 2018).

It should be highlighting that predictive analytics and intelligent data analysis are the key directions of the use of AI systems and technologies for decision-making support in modern companies (Serova, 2020a).

In the late 1970s and early 1980s many studies in the field of decision-making have been carried out based on experiments in the framework of static systems. However, over time, the conclusions of such studies were criticised and the need arose for conducting experiments in the field of decision-making in dynamic systems (DS). Studies

appeared in which subjects had to make decisions in experimental systems that included feedback, time-lag effects, and nonlinear behaviour. This has become possible thanks to the use of computer simulation models.

Currently, the so-called 'complexity economy' is gaining ground, which refutes the thesis about the desire to maximise utility when deciding by a person as an economic agent. The concept of economics as a complex adaptive system creates the need for a search and implementation of a new methodology that allows modelling processes. The behaviour of the system is formed from the relationships of many agents, each of which has certain behavioural features.

The major approaches (methods) in simulation modelling are SD, discrete event (DE), and agent-based (AB). SD and DE are traditional; AB is relatively new. There is also DS, but as a rule, it used to model and design 'physical' systems (Borshchev and Filippov, 2004, 2006; Karpov, 2005; Serova, 2012a, 2012b, 2012c; Krichevsky and Serova, 2016; Elberg and Tsygankov, 2017).

According to Serova (2012c), MAS and AB programming are the next step in the development of object-oriented programming and integrate the achievements in the field of AI, parallel computing, and knowledge-based systems. MAS are a self-organising decentralised network of agents (software objects). Each agent is autonomous and strives to achieve its own goal. Changing the purpose of one agent leads to the fact that others begin to adapt their behaviour and change connections.

Agent technologies are offering various types of agents, models of their behaviour and characteristics, ranges of architectures, and libraries of components.

MAS as systems of distributed AI have the following benefit (Serova, 2009):

- They speed up task fulfilment by parallelism and save the volume of data transmitted bypassing high-level partial solutions to other agents.
- They are flexible by using agents of various capacities to carry out a task dynamically in cooperation.
- They are reliable by-passing functions from agents unable to carry out a task to other ones (Serova, 2009).

Agents should have the following characteristics:

- The agent is identifiable, i.e., possesses a set of characteristic characteristics and principles that determine his behaviour and decision-making process; autonomous and can make decisions taking into account interaction with other agents while acting independently.
- The agent has a specific goal that affects his behaviour.
- Self-learning agent.

Every MAS consists of the following components (Serova, 2012c):

- Set of organisational units with a subset of agents and objects.
- Set of tasks.
- Environment, i.e., space where agents and objects exist.
- Set of relations between agents.
- Set of agent actions (e.g., operations on objects) (Serova, 2012c).

Another modelling approach is SD.

SD is a methodology for studying and modelling systems that are characterised by feedback cycles in complex mutual causal relationships of their parameters. This modelling approach was developed by Forrester in the 1950s (Forrester, 1961). Processes that occur in the real world in SD are characterised by the terms of stocks, flows between these stocks, and information that determines the value of these flows. SD abstract from individual objects and events and involves an 'aggregate' approach to processes. In SD, modelling is a set of interacting positive and negative feedbacks and delays, and the structure of the system determines its behaviour. After the model is built, it is possible to apply a simulation of numerical indicators, which in the form of a graph or table will represent the behaviour of the system. SD is a traditional approach, but it is successfully used for the decision of contemporary business tasks (Krichevsky and Serova, 2016; Saryazdi et al., 2020).

The main advantages of agent modelling compared to SD are:

- Modelling the behaviour of each of the agents.
- The possibility of adaptation.
- The presence of a mechanism for the formation of goals and a mechanism for interaction (communication and cooperation) of agents.

The main disadvantage of agent modelling compared to SD is that it is often difficult to develop a detailed agent model that takes into account all the cause and effect relationships that affect the agent's decision. Below a comparative table (Table 1) of two approaches to simulation is shown a brief but thorough description of the literature on the topic of the paper as well as recent research projects and their outcomes.

Table 1 Comparative characteristic of approaches to simulation

Parameter	*System dynamics*	*Agent modelling*
Basic model element	Feedback loop	Agent
Analysis area	System structure	Agent behaviour rules
Simulation level	Macro-level	Micro-level
Time	Continuous	Discrete
The core of modelling	Mathematics	Logic

Source: Katalevsky (2015)

Currently, there is different software in which it is possible to conduct a simulation. One of these programs is the AnyLogic. AnyLogic is a tool that covers the main areas of modelling: SD, agent modelling, etc.

The basic concept of AnyLogic is that the model is described by a set of interacting and simultaneously functioning activities, where the active object has its functioning and interaction with the environment.

AnyLogic graphical modelling environment is capable of supporting the following processes:

- Design.
- Development.
- Documentation of the model.
- Performance of computer experiments with the model.

3 Modelling of transportation, loading and unloading oil and oil product

There are various approaches to designing MAS. Yet, three levels can be found: conceptual description, initial design, and detailed design. At the first level, one should describe the organisational structure, goals, business processes, and information support underlying the next level's ontology. The next two levels are where all the elements described forming the organisational visualisation, i.e., the virtual world where agents act, using the ontology to achieve the goals and carry out the tasks set.

Because the behaviour of each of the participants in the processes of transportation, loading, and unloading operations with petroleum products is unique, agent modelling in AnyLogic was chosen to build the model and conduct a series of experiments.

There are several main stages to create a model (Table 2).

The science and business know examples of AnyLogic simulation models for the oil industry: for example, a system of sea oil transportation for Novoportovskoye field (PJSC Gazprom Neft). Oil from this field is transported via a 100-kilometre pipeline to the sea terminal located at Cape Kamenny (Guld of Ob), from where oil is transported to consumers on Arctic tankers. The main problem is difficult to ice conditions, which complicate navigation since most of the time the movement of ships in the Gulf of Ob takes place along a 500-kilometre canal through ice more than two meters thick. In the open part of the Kara Sea, drifting ice in some months even completely covers the seawater area. So, it was necessary to solve the following tasks:

- To develop a permanent oil export route: at this stage, it was required to determine the required number of vessels, as well as to understand whether the involvement of icebreakers would be required; it was also necessary to calculate the cost of fuel for tankers and the cost of freighting icebreakers.
- To design a temporary route for oil export: it was necessary to determine the throughput of the system during the period of use of light tankers with their subsequent replacement by tankers of larger capacity.
- To determine the required onshore oil terminal capacity: it was necessary to calculate the minimum onshore oil terminal volume, sufficient for the entire development period of the field.

Table 2 The main modelling stages

No.	*Stage*	*Result*
1	Analysis of the system	Understanding the functioning of the system, its structure, the processes taking place in it.
2	Formulation of the purpose of modelling the system	Compiling a list of tasks that need to be solved with the help of a future model, including a list of input and output parameters of the model, initial data, and criteria for completing the future study.
3	Development of the conceptual structure of the model	The structure of the model and its essential processes, a list of assumptions, a description of the control logic for subsystems.
4	Implementation of the model in a modelling environment	Implemented subsystems with characteristic behaviour, their parameters, and variables implemented logic and communications of the subsystems.
5	Implementation of the animation representation of the model	Animation representation of the model.
6	Verification of the correct implementation of the model	Verification of the correct display of the processes of the real system of the model for further analysis.
7	Calibration of the model	Fixing parameter values, coefficients of equations, and distributions of random variables, reflecting those situations for the analysis of which the model will be used.
8	Planning and conducting a computer experiment	Simulation results – graphs, tables, solving tasks.

To solve the above problems, the Krylov State Scientific Centre has developed a solution. As part of this solution, the modules of the GIS environment, shipbuilding calculations, and the AnyLogic logistics simulation model were combined, which made it possible to take into account the logic of transport operations, the characteristics of ships, and environmental conditions in the calculations.

The vessels in the system were presented as agents. At the same time, the model took into account the principles of interaction between tankers and icebreakers and the logic of choosing the speed of the vessel, depending on the fullness of the terminal, the severity of ice conditions, the position of other vessels, air temperature and other factors.

Figure 3 A visual image of the work of the oil supply chain model (in AnyLogic software) (see online version for colours)

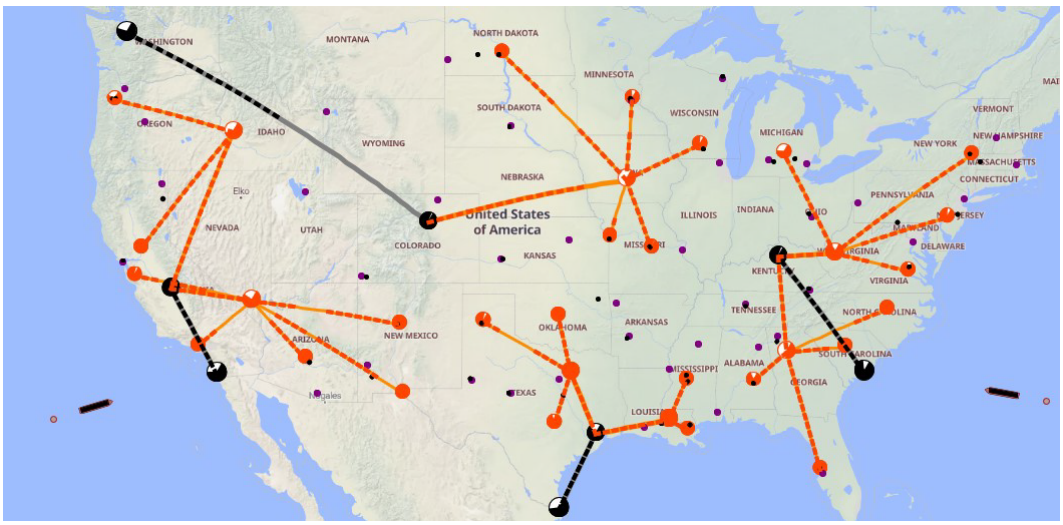


Figure 4 A visual image of the work of filling stations (with AnyLogic software) (see online version for colours)

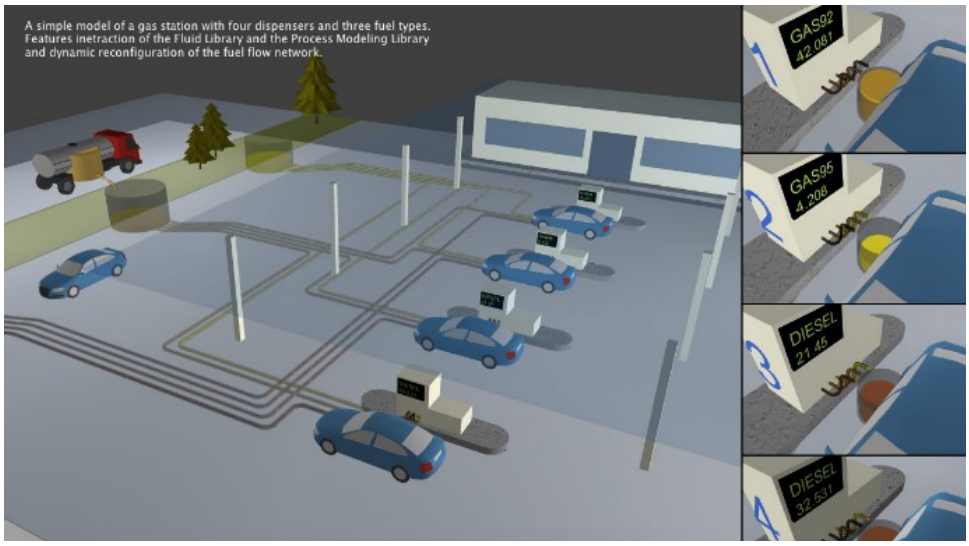


Figure 5 The architecture of the simulation model of the filling station (AnyLogic software) (see online version for colours)

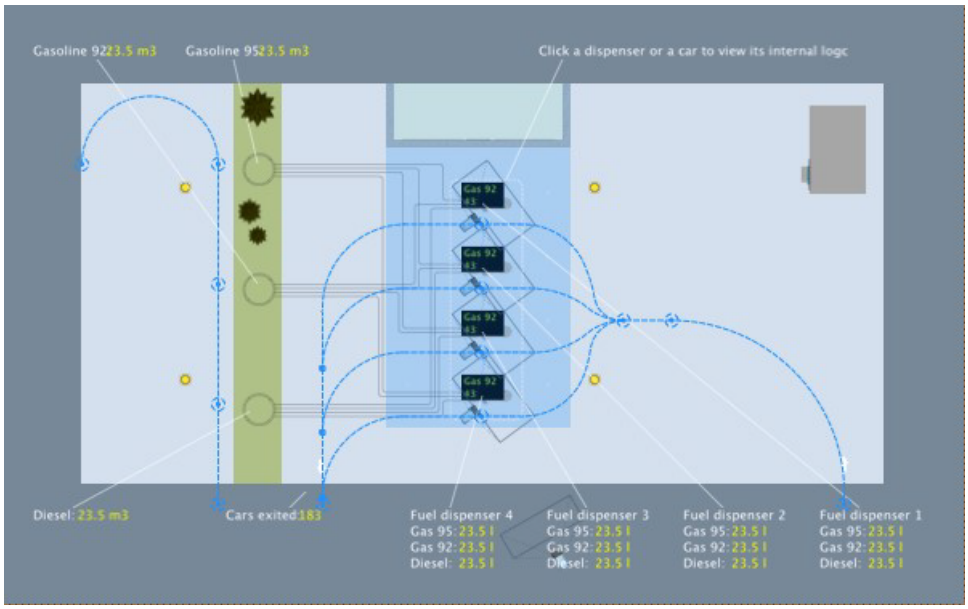
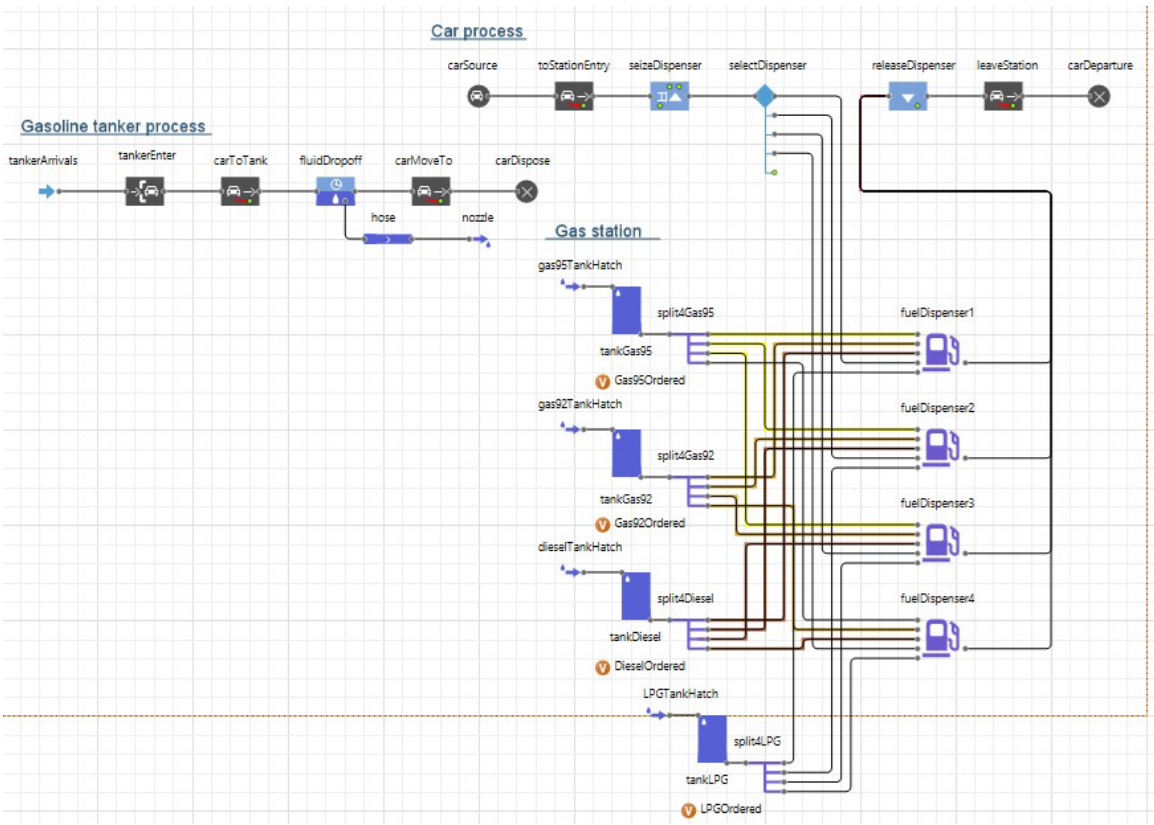


Figure 6 Flowchart (AnyLogic software)(see online version for colours)



As a result, the transport system of the Novoportovskoye field was designed taking into account the freezing of the ice canal, the dynamics of the movement of ships, and the filling of oil terminal facilities. The volume of the onshore terminal was determined, the required number of ice canals was estimated, the approximate dates of their construction were calculated, and the dates of icebreaker support for tankers were determined.

The next point of oil delivery is the refineries. So, there is another example of agent modelling in AnyLogic: the oil supply chain model. This model was developed in a GIS environment (on the example of the USA).

The agents are tankers, oil terminals (for oil), oil pipelines, oil refineries, oil product pipelines, oil terminals (for oil products), and fuel trucks. Each of them has its unique parameters (for example, capacity, speed, etc.)

In total the model consists of:

- four tankers
- four oil terminals (for oil)
- four oil pipelines
- four oil refineries
- 32 oil product pipelines
- 32 oil terminals (for oil products)
- 41 fuel trucks.

An example of this model is shown in Figure 3.

The model works as follows:

1 A tanker filled with oil ( ) approaches to the shore, unloads oil into the oil terminal ( ).

2 After that oil is transported by pipeline transport ( ) to the oil refinery ( ).

3 After that, the oil product is transported by oil product pipelines ( ) to the oil terminal (for products) ( ), from where the retailer ( ) transports the oil product to the final point of destination (for example, a filling station).

Examples of a sketch of AB simulation of a filling station with four columns that refuel vehicles with three types of fuel are shown in Figure 4 and Figure 5.

Agents in the above model are:

1 cars that come for refuelling

2 fuel trucks that come to drain the oil product

3 columns that load oil to cars.

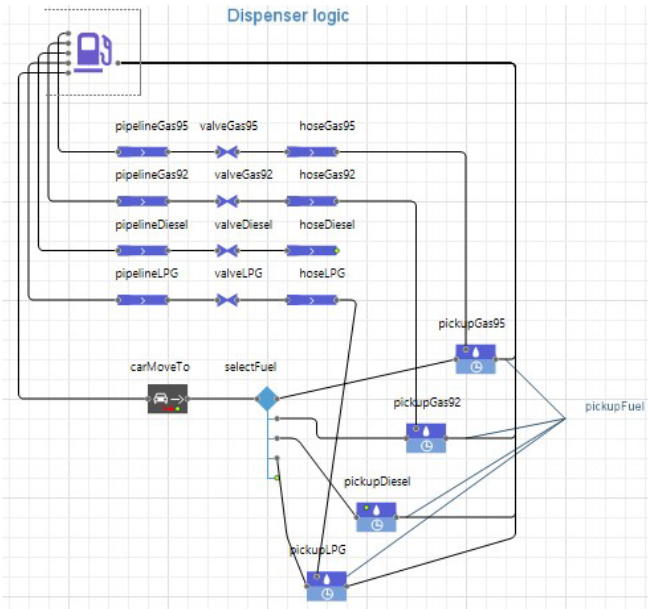
Initiatives of Russian oil companies in the sphere of corporate social responsibility include the sale of fuels that have a less negative impact on the environment. Modern gas stations refuel vehicles, including liquefied petroleum gas (propane, butane, and mixtures thereof), which are used as an alternative fuel for refuelling cars. The use of motor gas fuel is currently very important, because every year the domestic fleet, consisting of more than 34 million units of vehicles, emits 14 million tons of harmful substances

together with exhaust gases. And this makes 40% of the total industrial emissions into the atmosphere.

In this regard, an liquefied petroleum gas (LPG) tank was added to the existing model, that is a tank that receives liquefied petroleum gas, which is subsequently sent to the columns as an additional fourth type of fuel (to the existing RON-92, RON-95, diesel) (Figure 6).

The production chain originates from reservoirs filled with fuel trucks with specific fuel products. These tanks are filled when the content level drops to the minimum mark (object 'tankerArrivals'). Having connected the drain device to the receiving tank with the appropriate type of fuel, the fuel truck drains oil products ('fluidDropoff') into the tank. Subsequently, petroleum products enter the columns during the refuelling of cars. The column operation scheme is shown in Figure 7.

Figure 7 Detail flowchart (column) ('fuelDispenser') (see online version for colours)



When refuelling, the driver drives into the gas station site, drives up to one of the columns ('seizeDispencer'), selects the type of fuel ('selectDispenser'), refuels the car, and leaves the site. The object 'pickup [Fuel]' object at the moment of filling a full tank automatically stops the fuel filling procedure in a car.

4 Results and discussion

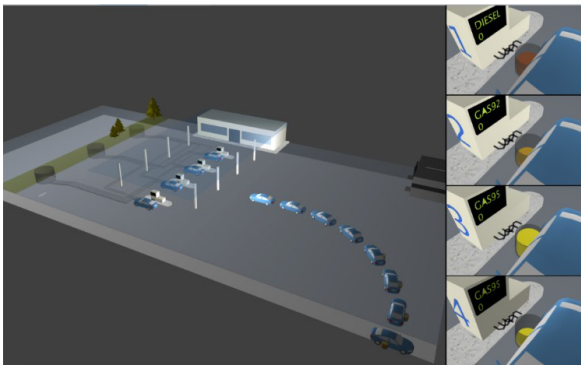
The model consists of the next agents:

- Tanks («tankGas95», «tankGas92, «tankDiesel»).
- Fuel dispenser.
- Gasoline tanker.
- Consumer (cars arriving at a gas station).

At the 8th stage (Table 2) of modelling (planning and conducting the experiments) – series of experiments on the filling station AB model were carried out to confirm that the model is working correctly.

1 Tank volume variation experiments. Agent: tanks («tankGas95», «tankGas92, «tankDiesel»):
   - In the case of an increase in the volume of tanks from 30 m^3 to 500 m^3, and in the case of an increase in the initial volume of filling the tanks to 400 m^3, the fuel truck does not arrive at the gas station for a much longer time in comparison with the initial set values.
   - In the case of a decrease in the volume of tanks from 30 m^3 to 4 m^3, and in the case of a decrease in the initial volume of filling the tanks to 0 m^3, a queue of cars is formed due to the lack of fuel at the gas station: in other words, there is a deficit of fuel at the gas station (Figure 8).

Figure 8 Tank volume variation experiment (see online version for colours)



2 Experiments with varying throughput of fuel supply pipeline, as well as fuelling nozzle. Agent: fuel dispenser.

   If the throughput of the fuel supply pipe and the fuelling nozzle decrease to 0, an error occurs so that the model is not working: indeed, why do we need an open idle gas station?

3 Experiments to change the rate of fuel discharged into the tank. Agent: gasoline tanker.

   In the case of an increasing rate of fuel discharge into the tank from 25 l/s to 150 l/s, it is obvious that the tank is filling faster: this significantly reduces the probability of a deficit at the gas station.

4 Experiment to change tank capacity. Agent: consumer (cars arriving at a gas station).

   Experiments included increasing of initially partly filled tank to the level of tank's capacity.

The main results of the AB model experiments are presented in Table 3.

Table 3 The main results of the experiments.

No.	*Experiment*	*Result*
1	Increasing the volume of tanks from 30 m^3 to 500 m^3, and increasing the initial volume of filling the tanks to 400 m^3.	The fuel truck does not arrive at the gas station for a much longer time in comparison with the initial set values.
2	Decreasing the volume of tanks from 30 m^3 to 4 m^3, a decreasing the initial volume of filling the tanks to 0 m^3.	There is a deficit of fuel at the gas station.
3	Increasing the rate of fuel discharge into the tank from 25 l/s to 150 l/s.	The probability of a deficit at the gas station significantly reduces.

5 Conclusions

The external environment in which economic systems operate and develop cannot be described with analytical expressions, and in such a situation, AI technologies, systems of distributed AI are the primary means of solving various economics and management tasks. Based on the multi-agent approach, an effective intelligent data analysis tool can be created. It is open and flexibly adaptable to the problem being solved, as well as can deeply be integrated with other systems.

MAS as systems of distributed AI are the concept that starts an era of network organisations with intellectual robots' collective interaction by offering to switch from powerful centralised systems to fully decentralised ones, with hierarchical structure replaced with network organisation, rigid bureaucratic 'from top to bottom' management (based on bosses' commands for subordinates) with negotiations, and planning with flexible agreements. As a result, production volumes, profitability, competitiveness, and mobility are growing. Major advantages of the multi-agent approach relate to the economic mechanisms of self-organisation and evolution that become powerful efficiency drivers and ensure enterprise's stable development and prosperity.

The application of AB modelling finds its niche in the field of solving problems of logistics and supply chain management. Thus, at present, for modelling processes of oil and oil products transportation, loading, and unloading, such types of simulation as AB modelling and SD can be used. Agent modelling involves the development of a behaviour model for each of the agents. SD, on the other hand, abstracts from individual objects and events and suggests an 'aggregate' approach to processes. As an example of the software used for logistics tasks simulation, AnyLogic is presented in this paper.

References

ABBYY IDC Research (2019) *Artificial Intelligence Market in Russia* [online] https://promo.abbyy.com/rs/446-REF-821/images/ABBYY%20IDC%20Research.pdf/ (accessed 25 February 2021).

AiRussia Online (2020) [online] https://www.airussia.online/ (accessed 25 February 2021).

Albright, S., Zappe, C. and Winston, W. (2011) *Data Analysis, Optimization, and Simulation Modeling*, Cengage Learning, Canada.

Borshchev, A. and Filippov, A. (2004) 'AnyLogic – multi-paradigm simulation for business, engineering and research', in *Proceedings of the 6th IIE Annual Simulation Solutions Conference*, Orlando, Florida, USA.

Borshchev, A. and Filippov, A. (2006) *From System Dynamics and Discrete Event to Practical Agent Based Modeling: Reasons, Techniques, Tools* [online] https://www.anylogic.com/upload/iblock/98a/98a7aebbe9e89099430f888f29a9715e.pdf (accessed 31 January 2021).

Dutton, T. (2018) An *Overview of National AI Strategy* [online] https://medium.com/politics-ai/an-overview-ofnational-ai-strategies-2a70ec6edfd (accessed 10 September 2020).

Elberg, M. and Tsygankov, N. (2017) *Imitation Modeling: Tutorial*, Siberia Federal University, Krasnoyarsk.

Forrester, J. (1961) *Industrial Dynamics*, MIT Press, Cambridge, MA.

Karpov, Y. (2005) 'System simulation modeling', *Introduction to modeling with AnyLogic*, BHV, St. Petersburg.

Katalevsky, D. (2015) *Fundamentals of Simulation and System Analysis in Management: A Tutorial,* 2nd ed., Delo, Moscow.

Krichevsky, M. and Serova, E. (2016) *Business Analysis and Decision Making Based on Data and Models*, Theory, Practice, Tools, Professional Literature, St. Petersburg.

Law, A. and Kelton, W. (2000) *Simulation Modelling and Analysis*, 3rd ed., McGraw-Hill.

Pidd, M. (2004) *Computer Simulation in Management Science*, 5th ed., Wiley.

Pospelov, D. (1998) 'Multi-agent systems – present and future', *Information Technologies and Computing Systems*, Vol. 1, pp.14–21.

SAP SE (2018) *The Future Customer Experience: Five Essential Trends* [online] https://www.sap.com/documents/2018/12/e2a78a40-2d7d-0010-87a3-c30de2ffd8ff.html/ (accessed 31 January 2021).

Saryazdi, A., Ghatari, A., Mashayekhi, A. and Hassanzadeh, A. (2020) 'The design of a system dynamic model of crowdfunding for support of new knowledge-based IT startups', *International Journal of Simulation and Process Modelling*, Vol. 15, No. 5, pp.454–474.

Serova, E. (2009) 'The role of multi-agent approach in building information infrastructure for a modern company and carrying out management tasks', *Intelligent Information and Engineering Systems // Supplement to International Journal Information Technologies and Knowledge*, Vol. 3, No. 13, pp.130–136.

Serova, E. (2012a) 'Distributed artificial intelligent systems for decision making support', in *Proceedings of the 26th Annual Conference of the British Academy of Management BAM 2012*, Cardiff Business School, Cardiff University, Cardiff, UK.

Serova, E. (2012b) 'Enterprise information systems of new generation', *The Electronic Journal Information Systems Evaluation*, Vol. 15, No. 1, pp.116–126 [online] https://academicpublishing.org/index.php/ejise/article/view/246/209 (accessed 31 January 2021).

Serova, E. (2012c) 'Intelligent decision support systems development based on modern modeling methods,' in *Proceedings of the 6th European Conference on Information Management and Evaluation ECIME 2012*, University College Cork, Ireland, pp.282–291.

Serova, E. (2020a) 'Is artificial intelligence essential to people, organization, and society?', in *Proceedings of the British Academy of Management Conference BAM 2020*, Oxford Abstracts, pp.1–8.

Serova, E. (2020b) 'MNEs' artificial intelligence strategies in the emergent economy', in *Conference Book Annual GSOM Emerging Market Conference 2020*, St. Petersburg University, Graduate School of Management, St. Petersburg, pp.94–97.

Serova, E. (2020c) *On the Issues of Artificial Intelligence Implementation: Advantages and Constraints' Becoming Greener – Digitalization in my Work*, LAB University of Applied Sciences, Lappeenranta, Finland, pp.49–53 [online] https://www.theseus.fi/bitstream/handle/10024/335985/LAB_2020_02.pdf?isAllowed=y&sequence=5 (accessed 31 January 2021).

Smith, C., McGuire, B., Huang, T. and Yang, G. (2006) *The History of Artificial Intelligence*, University of Washington.

Sterman, J. (2000) *Business Dynamics: Systems Thinking and Modeling for a Complex World*, McGraw Hill, Boston.

Wooldridge, M. (2002) *Introduction to MultiAgent Systems*, Wiley.

Yandex (2019) [online] https://yadi.sk/i/JywGWfSz3HKkpw (accessed 25 February 2021).

Yilmaz, L. (2020) 'Managing structural variability in agentbased models with feature coherence graphs', *International Journal of Simulation and Process Modelling*, Vol. 15, No. 5, pp.439–453.